Uniform almost-center domination with arbitrarily small loss

Partial result packet for arXiv:1610.02557

Run: `fa_banach_001`

Source question

The source paper is Timur Oikhberg and Pedro Tradacete, *Almost band preservers*, arXiv:1610.02557 [1]. It defines ε-band preserving operators and the ε-center $\varepsilon - \mathcal{Z}(X)$. The relevant passage is:

> "In general, if $T \in \varepsilon - \mathcal{Z}(X)$, then T is ε-BP. We do not know whether the converse implication holds in general."

this properties, we introduce:

Definition 4.2. *An operator $T : X \to X$ is in the ε-center (in short $T \in \varepsilon - \mathcal{Z}(X)$) if there exists $\lambda > 0$ such that for every $x \in \mathbf{B}(X)$, there is $z \in X$ with $\|z\| \leq \varepsilon$ such that*

$$|Tx| \leq \lambda|x| + z.$$

Note that $T \in 0 - \mathcal{Z}(X) = \mathcal{Z}(X)$ if and only if T is BP ([1]). Moreover, if T is BP, and S is arbitrary, then $T + S \in \|S\| - \mathcal{Z}(X)$. In general, if $T \in \varepsilon - \mathcal{Z}(X)$, then T is ε-BP. We do not know whether the converse implication holds in general. However, if $T \in ABP(\varepsilon)$, then $T \in \varepsilon - \mathcal{Z}(X)$. In Section 5 we will provide conditions for which every ε-BP operator is in $ABP(4\varepsilon)$, hence it also belongs to $4\varepsilon - \mathcal{Z}(X)$.

Note that $T \in \varepsilon - \mathcal{Z}(X)$ if and only if there is $\lambda \geq 0$ such that for every $x \in \mathbf{B}(X)$,

$$\|(|Tx| - \lambda|x|)_+\| \leq \varepsilon.$$

For $T \in \varepsilon - \mathcal{Z}(X)$, we define

$$\rho_\varepsilon(T) = \inf\{\lambda \geq 0 : \sup_{x \in \mathbf{B}(X)} \|(|Tx| - \lambda|x|)_+\| \leq \varepsilon\}.$$

The source paper's Theorem 2.10 shows that, for bounded T, the ε-BP condition implies a local domination statement: for each $x \in \mathrm{B}(X)$ and each $\varepsilon' > \varepsilon$, there is a scalar λ_x such that

$$\big\|(|Tx| - \lambda_x|x|)_+\big\| < \varepsilon'.$$

The open endpoint issue is whether λ_x can be chosen uniformly over $x \in \mathrm{B}(X)$ at the exact constant ε. The theorem below obtains such a uniform choice after replacing ε by $\varepsilon + \eta$.

Result

Theorem 1 (Uniform $\varepsilon + \eta$ center consequence)**.** *Let X be a Banach lattice, let $T \in B(X)$, and suppose that T is ε-band preserving. Then for every $\eta > 0$,*

$$T \in (\varepsilon + \eta) - \mathcal{Z}(X).$$

More quantitatively, if $M = \|T\| > 0$ and

$$\lambda \geq \frac{8M^2(M+1)}{\eta^2},$$

then for every $x \in \mathrm{B}(X)$,

$$\big\|(|Tx| - \lambda|x|)_+\big\| \leq \varepsilon + \eta.$$

If $T = 0$, the conclusion is trivial with $\lambda = 0$.

Proof intuition

The proof is a quantitative version of the source paper's proof of [1, Theorem 2.10]. That proof shows that if a norm-one functional x^* is disjoint from x, then x^* cannot see Tx by more than ε. We need the approximate form: if a positive norm-one functional almost annihilates $|x|$, say $\langle x^*, |x|\rangle \leq \alpha$, then $\langle x^*, |Tx|\rangle \leq \varepsilon + O(\sqrt{\alpha})$.

Now suppose $(|Tx| - \lambda|x|)_+$ has norm $> \varepsilon + \eta$. A positive supporting functional for this positive part may be restricted, inside the principal ideal generated by $|Tx| \vee |x|$, to the region where $|Tx| > \lambda|x|$. On that region it almost annihilates $|x|$, with $\alpha \leq \|T\|/\lambda$. Choosing λ of order η^{-2} contradicts the approximate disjoint-functional estimate.

A quantitative disjoint-functional lemma

Lemma 1. *Let $T \in B(X)$ be ε-BP and put $M = \|T\|$. Let $x \in \mathrm{B}(X)$ and $0 \leq x^* \in \mathrm{B}(X^*)$. If $\langle x^*, |x|\rangle \leq \alpha$, then for every $s > 0$,*

$$\langle x^*, |Tx|\rangle \leq \varepsilon + \frac{2\alpha}{s} + M(M+1)s.$$

Proof. Let $u = |Tx|$, $a = |x|$, and let I be the principal ideal generated by $u \vee a$. As in the proof of [1, Theorem 2.10], identify I with some $C(\Omega)$ by a lattice isomorphism $j : C(\Omega) \to I$ satisfying $j1 = u \vee a$. Then

$$\|j\| = \|u \vee a\| \leq \|u\| + \|a\| \leq M + 1.$$

Write $x = j(\phi)$ and $u = j(\phi')$. Since $0 \leq u \leq u \vee a$, we have $0 \leq \phi' \leq 1$.

Let μ be the positive measure on Ω induced by x^*, so that $\int f\, d\mu = \langle x^*, j(f)\rangle$ for $f \in C(\Omega)$. Then

$$\int |\phi|\, d\mu = \langle x^*, |x|\rangle \leq \alpha.$$

Choose $h \in C(\Omega)$, $0 \leq h \leq 1$, with $h = 1$ on $\{|\phi| \leq s/2\}$, $h = 0$ on $\{|\phi| \geq s\}$, and

$$1 - h \leq 2s^{-1}|\phi|.$$

Define

$$\psi = (\phi_+ - s1)_+ - (\phi_- - s1)_+, \qquad y = j(\psi),$$

and $y' = j(\phi' h)$. Then $|y| \leq |x|$, hence $\|y\| \leq 1$, while $\|x - y\| \leq (M+1)s$. Also $y \perp y'$, because ψ is supported where $|\phi| \geq s$ and h is supported where $|\phi| < s$.

The lower estimate from the source proof becomes quantitative:

$$
\begin{aligned}
\|u \wedge y'\| &\geq \langle x^*, y' \rangle \\
&= \int \phi' h \, d\mu \\
&= \langle x^*, u \rangle - \int \phi'(1-h) \, d\mu \\
&\geq \langle x^*, u \rangle - \frac{2}{s} \int |\phi| \, d\mu \\
&\geq \langle x^*, u \rangle - \frac{2\alpha}{s}.
\end{aligned}
$$

Since T is ε-BP and $y \perp y'$, we have $\| |Ty| \wedge y' \| \leq \varepsilon \|y\| \leq \varepsilon$. Moreover

$$
\| \, |Tx| - |Ty| \, \| \leq \|T(x-y)\| \leq M(M+1)s.
$$

Thus

$$
\begin{aligned}
\varepsilon &\geq \| |Ty| \wedge y' \| \\
&\geq \|u \wedge y'\| - \| \, |Tx| - |Ty| \, \| \\
&\geq \langle x^*, u \rangle - \frac{2\alpha}{s} - M(M+1)s.
\end{aligned}
$$

Rearranging proves the lemma. □

Proof of Theorem 1

Assume $M = \|T\| > 0$, fix $\eta > 0$, and choose

$$
\lambda \geq \frac{8M^2(M+1)}{\eta^2}.
$$

Let $x \in \mathrm{B}(X)$, and set $a = |x|$, $u = |Tx|$, and

$$
v = (u - \lambda a)_+.
$$

We prove $\|v\| \leq \varepsilon + \eta$.

Suppose, toward a contradiction, that $\|v\| > \varepsilon + \eta$. Let I be the principal ideal generated by $u \vee a$, represented as $C(\Omega)$ with u, a, and v corresponding to non-negative functions ϕ', ϕ, and $(\phi' - \lambda\phi)_+$. Choose a positive norm-one functional on X which sees v by more than $\varepsilon + \eta$, restrict it to I, and then restrict the representing measure to the open set

$$
A = \{\omega \in \Omega : \phi'(\omega) > \lambda\phi(\omega)\}.
$$

By the positive Hahn–Banach extension theorem for ideals, this restricted measure extends to a positive functional $x^* \in X^*$ with $\|x^*\| \leq 1$, and it still satisfies

$$
\langle x^*, v \rangle > \varepsilon + \eta.
$$

Since the measure is supported on A, we have $\phi \leq \lambda^{-1}\phi'$ on its support. Consequently

$$
\alpha := \langle x^*, a \rangle \leq \lambda^{-1} \langle x^*, u \rangle \leq \frac{M}{\lambda}.
$$

Also

$$\langle x^*, u \rangle \geq \langle x^*, v \rangle > \varepsilon + \eta.$$

Apply Lemma 1 with

$$s = \frac{\eta}{2M(M+1)}.$$

The choice of λ gives

$$\frac{2\alpha}{s} \leq \frac{2(M/\lambda)}{s} \leq \frac{\eta}{2}, \qquad M(M+1)s = \frac{\eta}{2}.$$

Therefore Lemma 1 yields

$$\langle x^*, u \rangle \leq \varepsilon + \eta,$$

contradicting $\langle x^*, u \rangle > \varepsilon + \eta$. Hence

$$\big\|(|Tx| - \lambda|x|)_+\big\| \leq \varepsilon + \eta$$

for every $x \in \mathrm{B}(X)$, as required.

Verification notes and limitations

This is not the exact endpoint converse asked in the source paper. It proves that the endpoint is the only possible obstruction:

$$\inf_{\lambda \geq 0} \sup_{x \in \mathrm{B}(X)} \big\|(|Tx| - \lambda|x|)_+\big\| \leq \varepsilon.$$

What remains unknown from this argument is whether the infimum is always attained at a finite λ. The bound here grows like η^{-2}, so it gives no endpoint compactness as $\eta \downarrow 0$.

The proof was checked against the source's proof of Theorem 2.10; the only new ingredient is the quantitative estimate in Lemma 1. Local run-index searches found the prior exact $C_0(K)$/AL-space packet for this target but no packet for the global $\varepsilon + \eta$ statement.

References

[1] T. Oikhberg and P. Tradacete, *Almost band preservers*, arXiv:1610.02557, 2016. https://arxiv.org/abs/1610.02557